



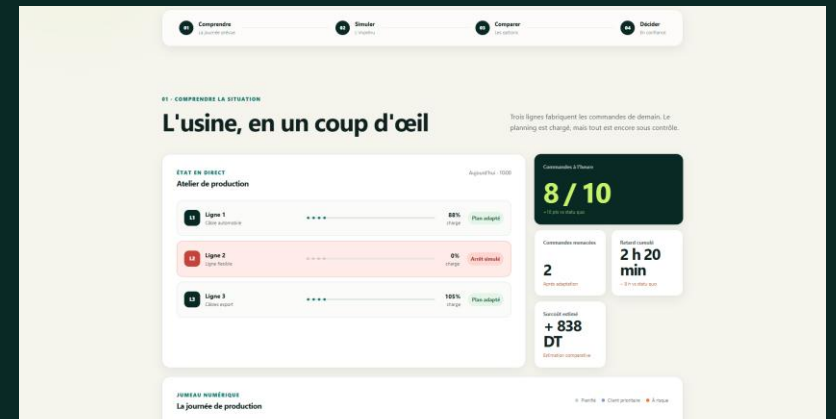
CableTwin

When a line stops, see the consequences
before changing production.

The Waze of the cable factory

Working offline prototype · Fictional Tunisian factory · Fully synthetic data

CableTwin · SUPCOM · Oussama Akir · Fully synthetic data



THE PROBLEM

At 10:00, one stoppage exposes 3 orders and 620 minutes of delay

Which orders should be moved, delayed or protected — and what will each choice do to service, throughput and the workshop?

Eligible lines

Durations

Due times

No overlap

Planning horizon

The active schedule becomes infeasible the moment the line stops: two orders overlap the stop, one downstream order is exposed, and every alternative must respect real constraints. The decision owner is the production manager.

The manager needs a feasible choice, not another dashboard.

LINE 2 UNAVAILABLE

10:00 – 14:00

ORDERS EXPOSED

3

ON TIME WITHOUT ADAPTATION

7 / 10

PROJECTED CUMULATIVE DELAY

620 min

CableTwin turns one disruption into three feasible routes

1 Detect the stop

2 Reveal exposed orders and KPI

3 Compare Service · Cost · Stability

4 Preview, approve and record

Like Waze after a road closes:

same destination, different routes and visible trade-offs — the driver still decides.

One incident → three comparable recovery schedules → one traceable human decision

Read-only decision support · No failure prediction · No machine control

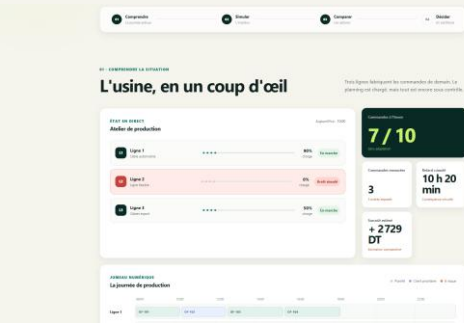
One end-to-end loop connects the incident to an auditable decision

01 — Understand



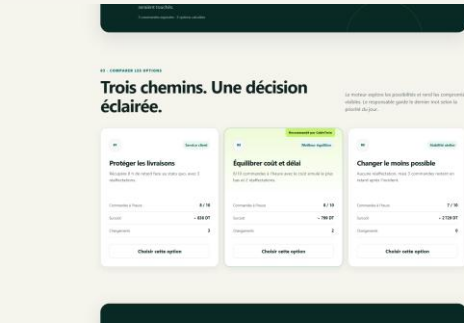
3 lines · 10/10 on time

02 — Simulate



Line 2 stops · 3 exposed

03 — Compare



3 feasible routes · same KPI

04 — Decide



Manager approves · audit at 10:07

The same executable path connects data, incident analysis, planning, Gantt preview, approval, audit and reset.

Service cuts projected delay from 620 to 140 minutes

620 → 140 min

projected cumulative delay

188 → 216 km

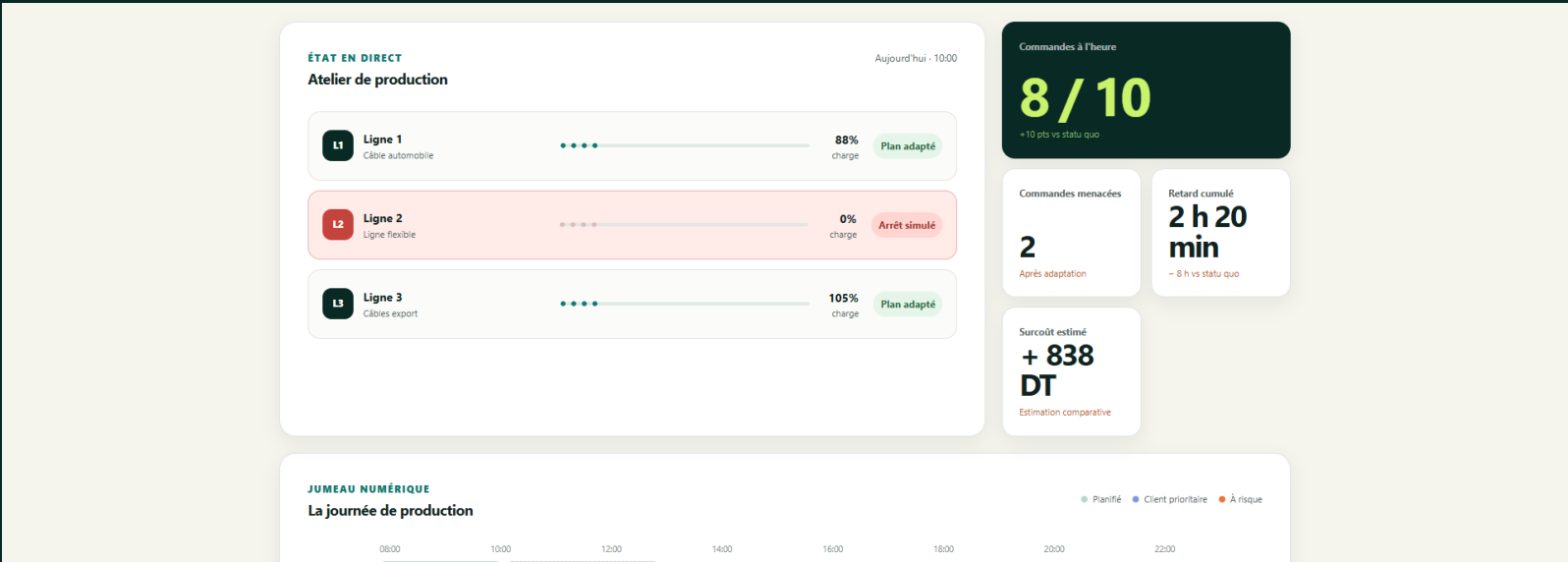
completed by 18:00 · +14.9%

180 → 30 min

overtime · -83.3%

7/10 → 8/10

orders on time · 3 line moves



Stability

620 min · 188 km · 0 moves

Service

140 min · 216 km · 3 moves

Cost

170 min · 216 km · 2 moves

Fixed synthetic scenario · calculated by the working prototype · not factory ROI

The AI is the planning decision, not the dashboard

LIVE SYMBOLIC PLANNER

Decision variables

line assignment · sequence · start / end

Hard constraints

eligibility · duration · downtime · no overlap · horizon

Business policies

Service · Cost · Stability (lexicographic objectives)

INDEPENDENT BOUNDED ORACLE

17,856 candidate schedules evaluated

10,440 feasible complete schedules

3/3 displayed routes = unique policy optimum

9/9 automated checks pass

A 9/10 route exists, but creates 230 min of total delay; Service chooses 8/10 with only 140 min because priority-weighted lateness matters. Exact for the encoded bounded demo — not factory-scale optimization.

Simulation proves the mechanism; a shadow pilot will prove value

PROVED IN DEMO

Same incident
Same KPI definitions
Reproducible before/after
calculations

5–10 historical stoppages

Same information cutoff as the real decision

Planner-validated constraints

CableTwin vs actual historical baseline

Time, delay, output, overtime, moves, usability

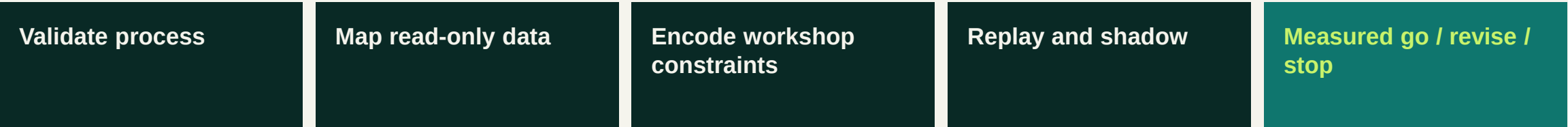
PROPOSED SUCCESS GATES — targets to agree, not achieved claims

30% faster assessment · 20% lower delay · 0 hard-constraint violations · ≥70% planner usability · 100% traceability

Ends in Go · Revise · Stop

Start read-only in one workshop; scale only after measured proof

User: **Production manager / planner** Buyer: **Plant or operations director** Pilot: **4–6 weeks · read-only**



Manual / spreadsheet	familiar but hard to compare and trace
ERP · MES · APS	system of record / broad planning
High-fidelity twin	rich but integration-heavy
CableTwin	focused incident layer · 3 explainable routes · human approval

No prediction · no autonomous control · no write-back · no price or ROI claim

READY FOR PHASE 3

CableTwin is ready for Phase 3

25% · Business process & impact

One incident-rescheduling loop · delay, overtime and output measured

20% · Core AI

Constraint planning · 17,856-candidate independent bounded certificate

20% · Working prototype

Offline incident → routes → Gantt → approval → audit → reset · 9/9 checks

20% · Business viability

Named user and buyer · 4–6-week read-only pilot · go/no-go gates

15% · Pitch & clarity

One Waze metaphor · three routes · one human decision

Select CableTwin for the final pitch and see the complete live decision in 90 seconds.